INCH-POUND
MIL-PRF-3098/137D
28 July 2011
SUPERSEDING
MIL-PRF-3098/137C
27 July 2005

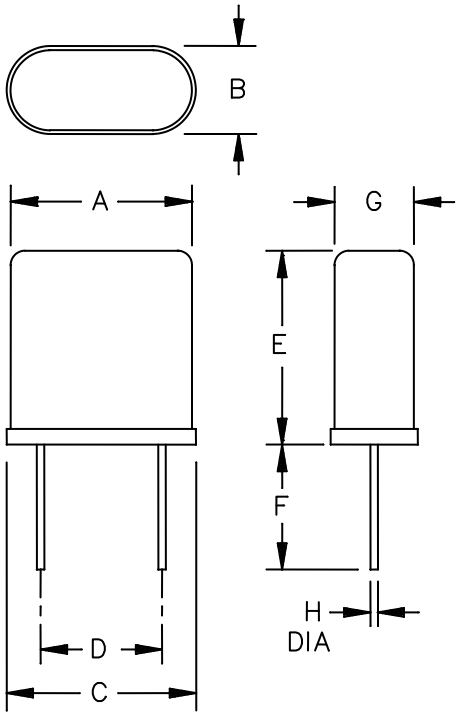
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR157/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 0.8 MHz to 20 MHz; fundamental; noncontrolled; series resonance.



Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.725	---	18.42
B	---	.352	---	8.94
C	---	.757	---	19.23
D	.478	.494	12.14	12.55
E	---	.775	---	19.69
F	---	.50	---	12.7
G	---	.317	---	8.05
H	.028	.032	0.71	0.81

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The wire leads shall be tinned, suitable for soldering.
4. Marking shall be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR157/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 MHz to 20 MHz, inclusive.

Frequency tolerance: ±50 ppm.

Equivalent resistance: See table I.

TABLE I. Equivalent resistance.

Frequency range, inclusive. MHz.	Maximum resistance Ohms
0.80 to 0.85	520
0.85+ to 0.90	480
0.90+ to 1.00	440
1.00+ to 1.12	400
1.12+ to 1.25	380
1.25+ to 1.37	340
1.37+ to 1.50	300
1.50+ to 1.62	280
1.62+ to 1.75	250
1.75+ to 1.87	220
1.87+ to 2.00	190
2.00+ to 2.12	170
2.12+ to 2.25	150
2.25+ to 2.60	130
2.60+ to 3.00	90
3.00+ to 3.40	70
3.40+ to 3.75	52
3.75+ to 4.00	45
4.00+ to 5.00	37
5.00+ to 7.00	25
7.00+ to 10.00	20
10.00+ to 15.00	18
15.00+ to 20.00	15

Mode of oscillation: Fundamental.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Capacitance, shunt: 7.0 pF, maximum.

Shock (specified pulse):	0.8 MHz to 2.0 MHz	2.0 MHz+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

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Vibration: Method 201, MIL-STD-202.	0.8 MHz to 2.0 MHz	2.0 MHz+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

Temperature cycling:	0.8 MHz to 2.0 MHz	2.0 MHz+ to 20.0 MHz
Frequency change permitted:	±10 ppm	±5 ppm
Equivalent resistance change permitted:	±15 percent	±10 percent

Aging:

Frequency change permitted: ±5 ppm.

Part or Identifying Number (PIN): For Product Level B, PIN shall be CR157/U or CR157B/U (followed by specified frequency). For Product Level S, PIN shall be CR157S/U (followed by specified frequency, see 1.2 of MIL-PRF-3098.

Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 99
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, MC, SH
Air Force - 19, 84

Preparing activity:
Army - CR

Agent:
DLA - CC

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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